

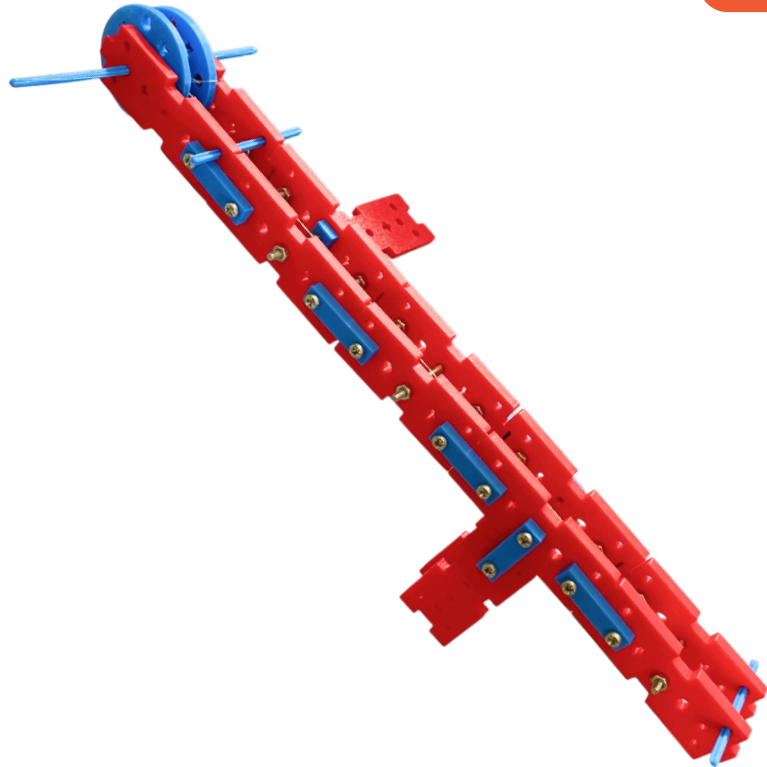
level 1

fishing rod

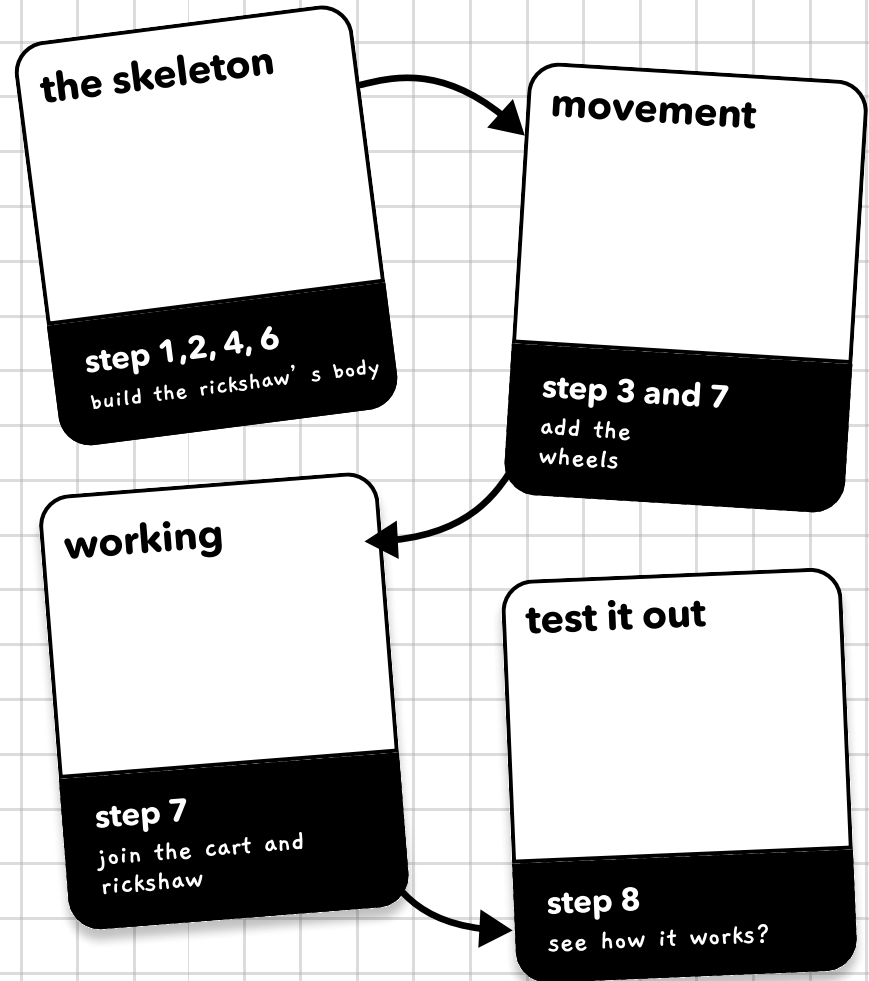
i help to catch fish by using less effort! My long rod acts like a lever, and your hand as the fulcrum. When you lift my handle, my other end pulls up the fish – showing how levers make lifting easier, even in water!

no. of sessions: 4
simple machine:
lever & pulley

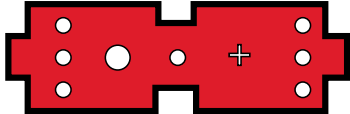
oh.



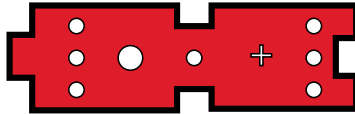
what's the goal?



materials needed



m-m beam - x1



m-f beam - x4



l clamp - x4

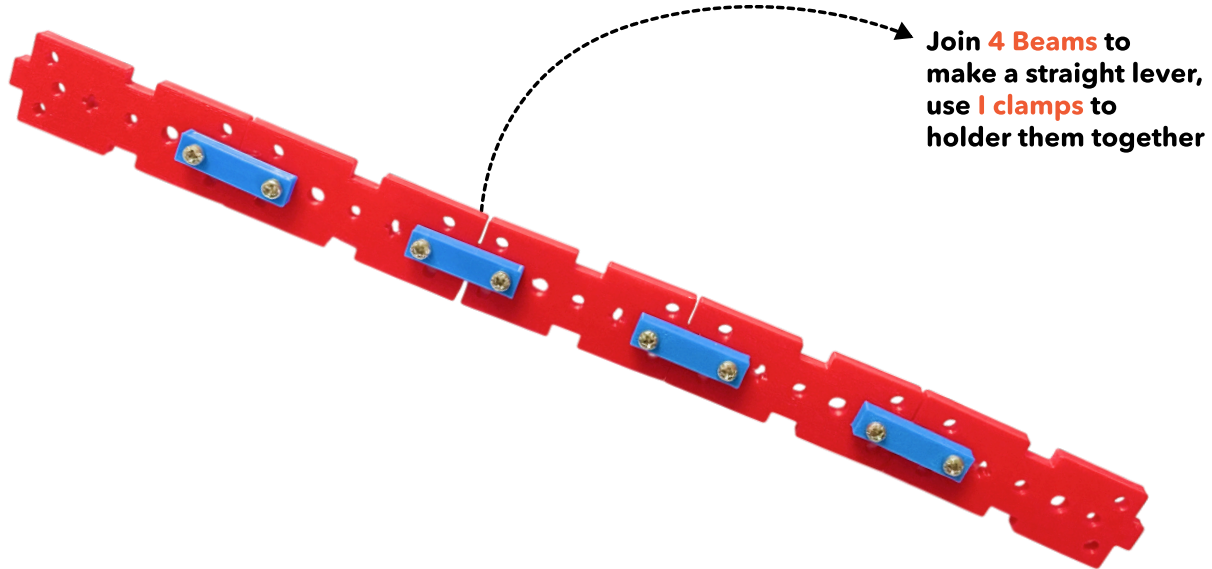


small screw - x8

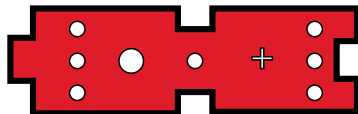


nut - x8

step 1: assemble the back chassis



materials needed



m-f beam - x1



I clamp - x1

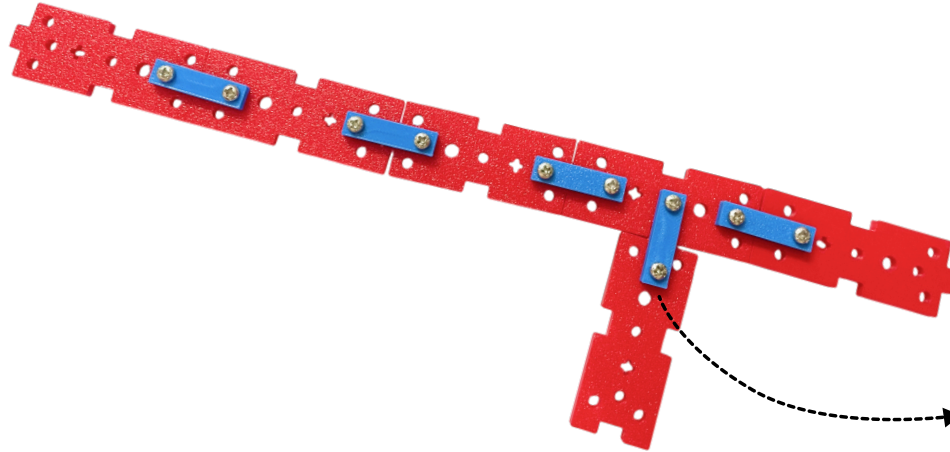


small screw - x2



nut - x2

step 2: add the handle to the lever



Use a **M-F beam** as handle to support the lever.

Join the beam using an **I clamp** as shown.



Repeat the steps 1 and 2 to make another set.

materials needed

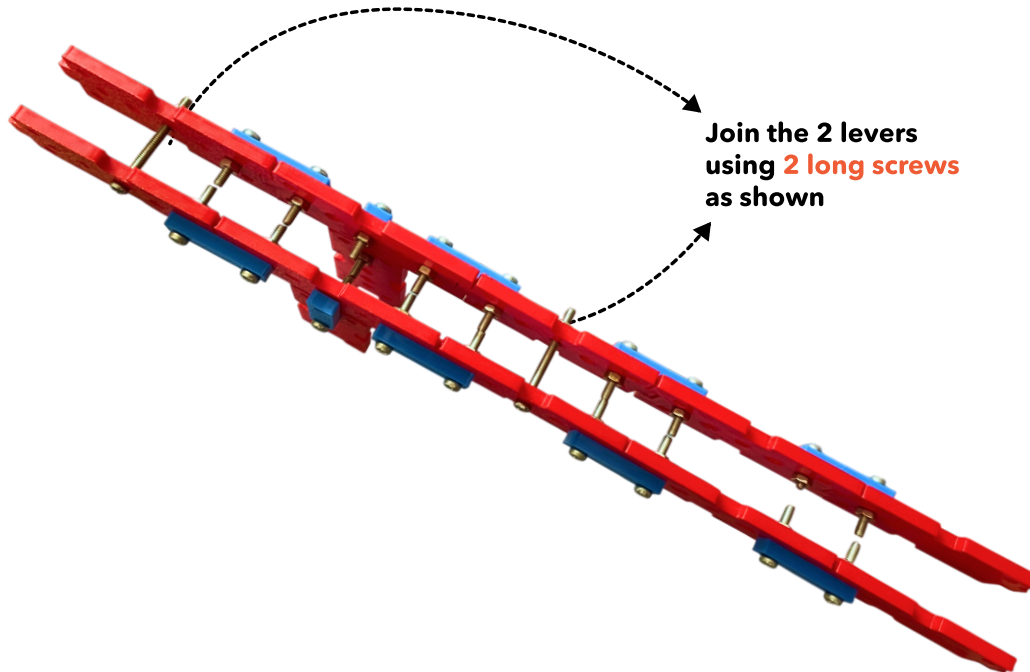


long screw - x2



nut - x2

step 3: join the two levers together.



Join the 2 levers
using **2 long screws**
as shown

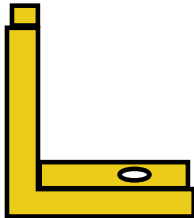
materials needed



long screw - x1

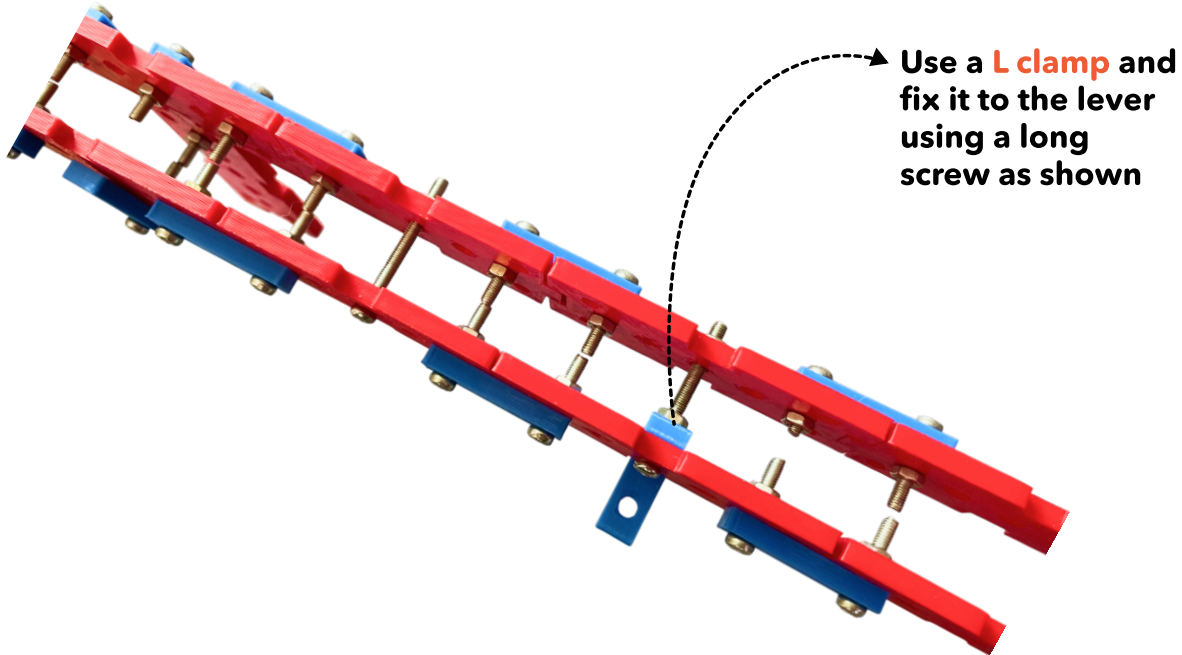


nut - x1

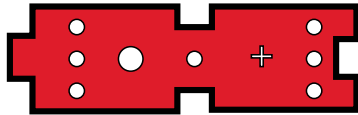


L clamp - x1

step 4: assemble the 2nd handle support



materials needed



m-f beam - x1

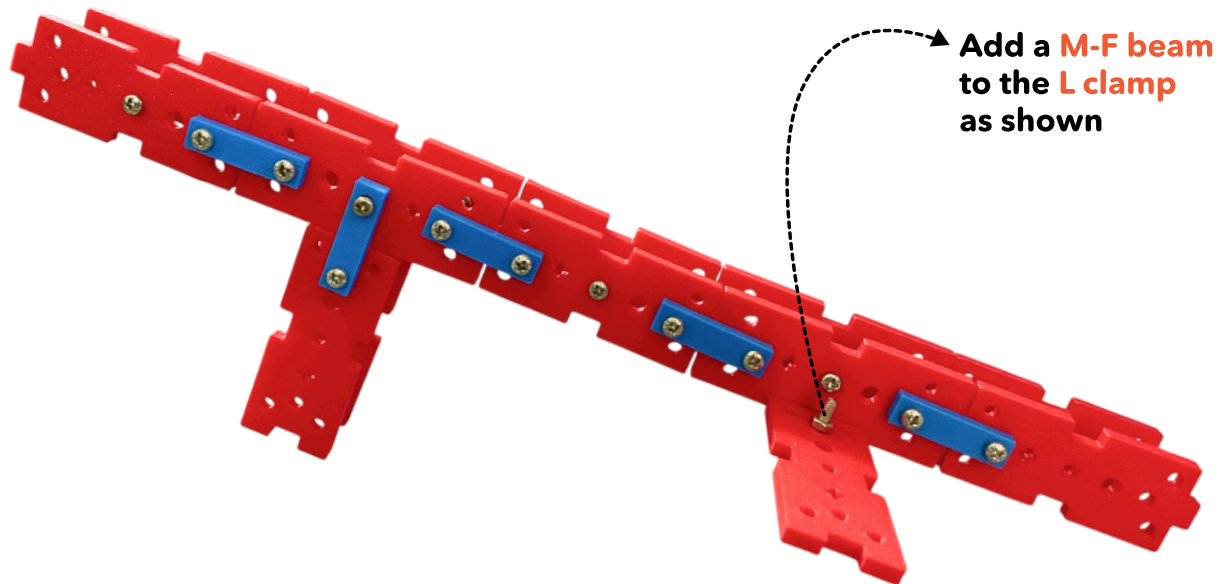


small screw - x1



nut - x1

step 5: fix the second handle

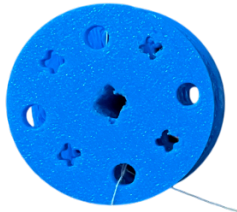


materials needed

long axle x1

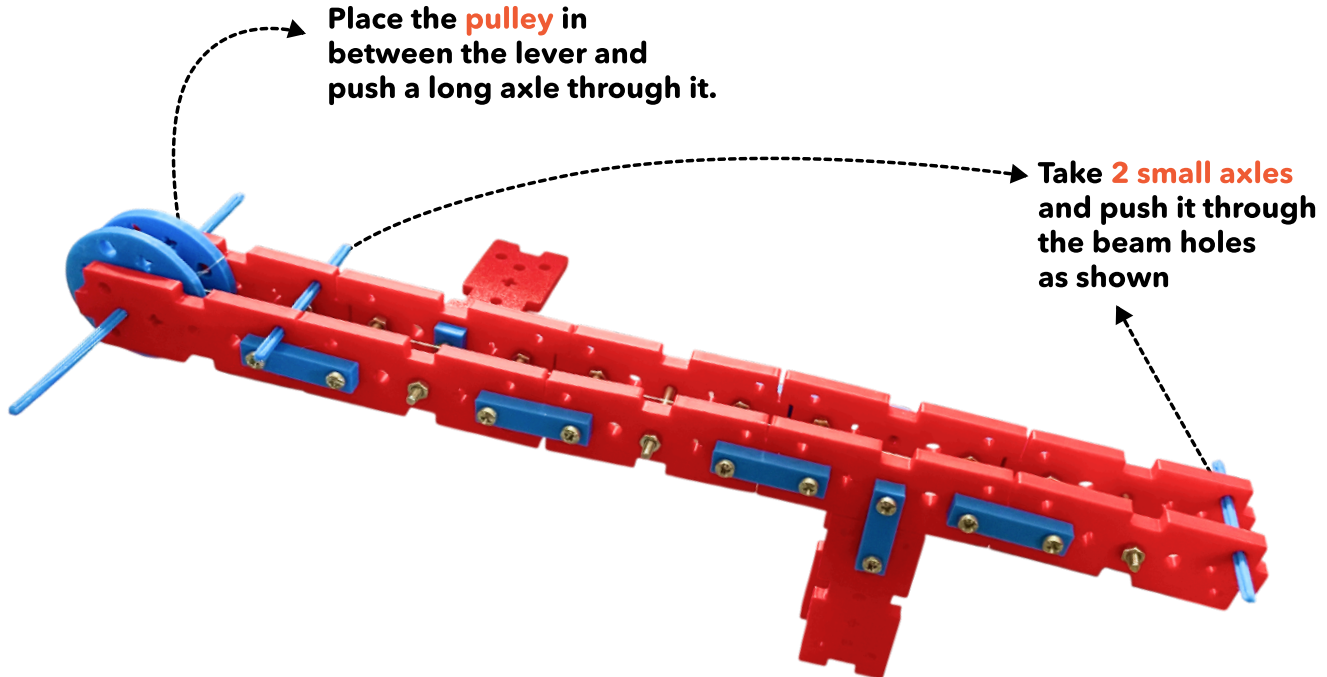


small axle x2



pulley - x1

step 6: assemble the pulley system



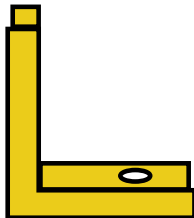
materials needed



washers - x6



rope x1



L clamp - x1

step 7: assemble the pulley handle

Finally tie a rope to pulley and pass the rope around the axles as shown.

Tie a L clamp as a hook to the other end of the rope.

